Core-Shell Catalyst Pellets: Theoretical Background, Preparation, and Experimental Validation

Ronny Tobias Zimmermann[1], *Kai Sundmacher*[1,2]

[1] *Max Planck Institute for Dynamics of Complex Technical Systems, Department for Process Systems Engineering, Sandtorstraße 1, 39106 Magdeburg*

[2] *Otto von Guericke University Magdeburg, Lehrstuhl für Systemverfahrenstechnik, Universitätsplatz 2, 39106 Magdeburg*

Fixed-bed reactors are generally designed to operate in a single steady-state over long periods of time. With the transition to renewable energy sources, which are generally more volatile than conventional feedstocks, there is growing interest in load-flexible operation of fixed-bed reactors, e.g. as part of the Power-to-X concept. Load-flexible reactors are expected to be able to be operated safely and efficiently at several operating points in order to minimise the costs of intermediate storage of the reactants. However, this approach requires a detailed understanding of the underlying physical and chemical phenomena.

The use of uniformly active catalyst pellets in fixed bed reactors can lead to positive feedback between the heat release and the temperature rise due to the non-linear Arrhenius temperature dependence of the reaction rate, which is referred to as thermal runaway. By combining analytical equations with the ‘practical design criterion’, we will show in this contribution that the intuitively obvious approach of diluting uniform active catalyst pellets with inert pellets cannot necessarily prevent thermal runaway. The same applies to reducing the activity of the catalyst pellets. Furthermore, these measures also have a detrimental effect on the product yield of the reactor [1].

Instead, we propose the use of core-shell catalyst pellets. In core-shell catalyst pellets, the catalytically active material is separated from the gas bulk by an inert layer, as is the case for example with ‘egg white’, ‘egg yolk’ or pellets with Dirac-δ distribution of the active material. Reactants must first diffuse through this inert layer before they can react. As diffusion processes, especially Knudsen diffusion, have a comparatively low temperature dependence, the mass transport through the inert shell becomes rate-determining above a certain temperature. If this is the case, the positive feedback between heat release and temperature increase is interrupted and the heat can be dissipated from the reactor tubes in a controlled manner by wall cooling [2,3].

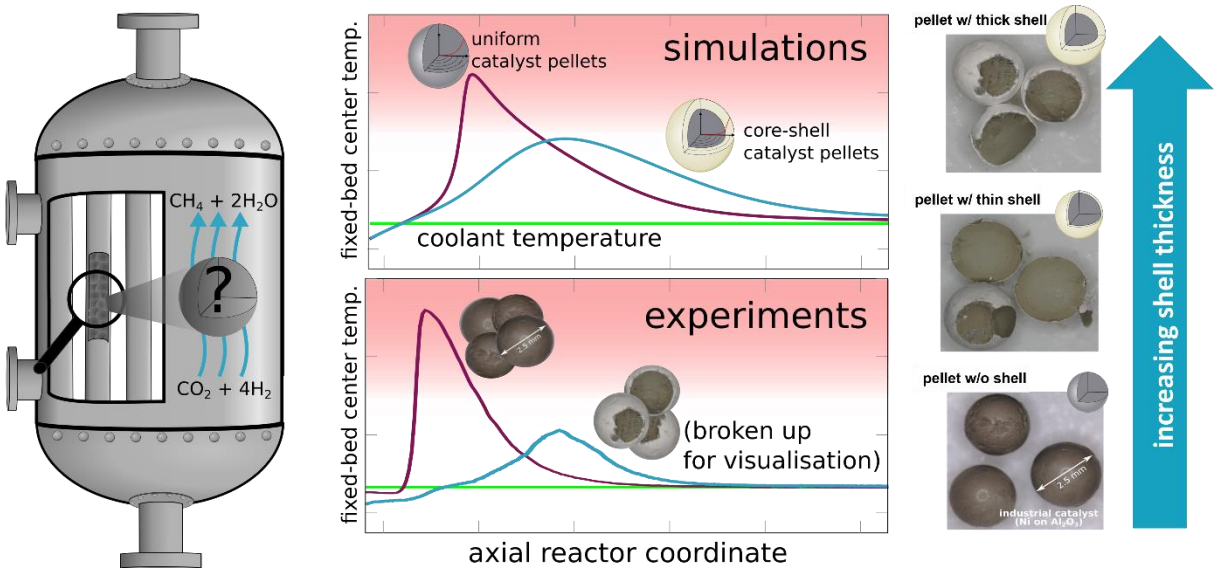

Figure 1: Computer-aided (top) and experimental (bottom) comparison of the temperature profiles in fixed-bed reactor by employing uniform catalyst pellets and corresponding core-shell catalyst pellets. The results were obtained using a fibre-optic temperature sensor in an oil-cooled single-tube fixed-bed reactor with a length of 2 m and a diameter of 2 cm. [4]

On this basis, we will show that core-shell catalyst pellets can be produced, by coating commercially available catalyst pellets in the fluidised bed. This process is industrially established, enables the production of large quantities of core-shell catalyst pellets and allows the properties of the inert shell to be tailored to the requirements of the respective chemical process. The shell is made of inexpensive materials such as aluminum oxide [2,3].

One prominent process engineering advantage of this concept is that core-shell catalyst pellets can be used directly in existing fixed-bed reactors without the need for additional equipment. They can therefore be integrated into existing plants. First experimental results with ‘egg yolk’ core-shell pellets confirm our simulation results using the example of the CO_2 methanation, a shown in Fig. 1 [4].

References

[1] Zimmermann R. T., Bremer J., Sundmacher K.: Optimal Catalyst Particle Design for Flexible Fixed-Bed CO_2 Methanation Reactors, *Chem. Eng. J.* (2020) **387**, 123704

[2] Zimmermann R. T., Sundmacher K.: Core-Shell Catalyst Pellets for Reaction Heat Release Control in Fixed-Bed Reactors, *Ind. Eng. Chem. Res.* (2024) **63.17**, 7556-7564

[3] Zimmermann, R. T. et al.: Core-Shell Catalyst Pellets for Effective Reaction Heat Management, *Chem. Eng. J.* (2023) **457**, 140921.

[4] A. Geschke, R. T. Zimmermann, J. Bremer, K. Sundmacher, Core-Shell Catalyst Pellets for CO_2 Methanation in a Pilot-Scale Fixed-Bed Reactor, *CIT*, (accepted)